

# SIH26151

## The Competitive Landscape

---

Every public GitHub repository working on Dark Web Threat Actor De-anonymization — scraped, compared, and ranked against NETRA-X.

**16 competing implementations found.** Each was fetched and read directly rather than inferred from its name. Where a repository states no tests, no metrics, or no scoring method, that absence is recorded as a finding — it is the most common weakness in the field, and the one NETRA-X is best positioned to exploit.

## 1.2 What the whole field gets wrong

### THE GAP NOBODY ELSE HAS CLOSED

Across all 16 repositories, **not one mentions handling correlated evidence**. The scoring methods found were:

- **Noisy-OR** — assumes conditional independence by construction
- **Weighted linear sums** — assume independence and additionally saturate
- **Fellegi–Sunter** (DRISHTI) — the right framework, but no  $\lambda$  or dependence-group handling described
- **Unstated** — the majority

In this domain the assumption is demonstrably false. A reused Bitcoin address and the co-spend cluster containing that address are **one operational mistake observed twice**. Every competitor counts it twice. NETRA-X counts it once plus a quarter, via  $\lambda = 0.25$  within a declared dependence group.

# Threat Profiles

Tier 1 — Serious competition

Working systems with published methodology and evidence of engineering discipline

**sharan112212/darkweb**

Python · Streamlit · 238 tests · ~24 modules

Scoring

Noisy-OR,  $P = 1 - \prod(1 - w_i)$ . Four categories (K cryptographic, I infrastructure, B behavioural, S semantic). Caps: text alone  $\leq 0.20$ , lone infra/behavioural  $\leq 0.65$ . Five tiers with  $\pm 0.03$  hysteresis.

Stack

FastAPI, Streamlit, PostgreSQL 16, Neo4j 5, MinIO, Nginx, Tor proxy, Docker Compose

ML

MiniLM sentence-transformers — a genuinely pretrained model. 50+ stylometric features via pystylometry.

Strengths

238 tests. OnionScan integrated. RBAC across 4 levels. SHA-256 hash-chained audit log. Tor in-repo.

Weaknesses

Noisy-OR saturates toward 1.0 as weak signals accumulate. Weights are asserted, not decomposable. No negative evidence described. Streamlit UI is a prototyping tool, not a product surface.

Beat them by

Dependence discounting and uncapped contradictions — both structurally absent from Noisy-OR. Our Next.js UI outclasses Streamlit on a demo video.

**bhedanikhilkumar-code/SENTINEL-X**

TypeScript + Python · 41 commits · 27 endpoints · 9 containers

Scoring

Multi-vector confidence  $C_{total}$ . Method not published in detail.

Stack

FastAPI async, Python 3.12, PostgreSQL 16, Neo4j 5.18, Redis 7, ChromaDB, Celery, React 18 + Vite + Cytoscape + Leaflet, Tor proxy

ML

SBERT 384-D embeddings for stylometry. UTXO peel-chain and mixer detection.

Strengths

Closest architectural match to NETRA-X, and ahead on ops: **RBAC enforced (4 roles)**, **CI/CD in .github/workflows**, Celery task queue, 9-container compose, geospatial layer.

Weaknesses

Claims "court-admissible documentation via Section 65B compliance" — a legal claim no hackathon prototype can support, and an obvious line of attack for a judge. No test count published. Synthetic data only. Two hardcoded demo targets (0.912, 0.760).

Beat them by

Methodological transparency. Their fusion is a black box; ours shows every item's contribution, discount and cap. Also press the Section 65B claim — it will not survive scrutiny.

**AnuAnsh17/DRISHTI**

Python · updated within 24h — actively developed

Scoring

Fellegi–Sunter for entity disambiguation, plus Dempster–Shafer / Bayesian confidence modelling. The same theoretical foundation as NETRA-X.

Stack

Plugin-based collectors, temporal graph modelling, provenance tracking with source-reliability scoring

Strengths

Only other team citing Fellegi–Sunter. Neutralises "we use the forensic likelihood-ratio standard" as a unique claim. Research directory suggests genuine literature grounding.

Weaknesses

**No test count. No metrics. No numbers at all.** Their README describes a synthetic evaluation lab measuring precision, recall and false positives — with the results left blank. Framework named, results absent.

Beat them by

Running numbers. `python -m bench.report` outputs ROC-AUC 0.9028, ECE 0.0044, FAR 0.00%, Recall@10 100%, reproducibly at seed 42. Also: they cite the framework; we implement the correction ( $\lambda$ ) that the framework alone does not give you.

Tier 2 — Credible but thin

Driven-Ansh/sih26151-traceforge

FastAPI + React · 5 commits · 18+ endpoints

- Scoring**
- Calibrated linear weighting — PGP 0.25, UTXO 0.20, stylometry 0.18, activity heatmap 0.15, infrastructure 0.12, temporal 0.06, trust network 0.05; minus two contradiction deductions; × 0.95 source reliability.
- Claimed metrics**
- 94.2% precision, 91.5% recall, 92.8% F1, +18.6% over stylometry alone — across **5 demo scenarios**.
- Assessment**
- Nearly identical stack to ours (FastAPI, React, Cytoscape, Render). **They do have contradiction deductions** — the only competitor that does. But the weights are asserted with no derivation, and precision/recall on five hand-built cases is not a benchmark. Quoting 94.2% from n = 5 is a number a judge can dismantle.

rohantiwari9573/SIH-26 "Argus" · hocuspocus07/argus

Python · both active within the last week

- Assessment**
- Two separate projects sharing a name. Both actively developed, neither publishes methodology, tests or metrics. Unknown quantities — worth re-checking before the final round, since activity suggests they are still building.

Tier 3 — Not competitive

Prototypes, scaffolds, or abandoned

| Project                                               | Evidence   | Why it is not a threat                                                                                                        |
|-------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------|
| kalees-bytes/SIH26151                                 | 2 commits  | Flat weighted sum <code>0.30·infra + 0.40·identity + 0.30·stylometry</code> . No tests, no CI, no metrics. Flask + Streamlit. |
| xarjunpatil/SIH26151                                  | 1 commit   | Single-file FastAPI app plus a glassmorphic HTML dashboard. Scoring described only as "AI anomaly scoring".                   |
| siddharthreddy432/CipherForge                         | 2 commits  | React + Express with an <b>in-memory graph simulation</b> — no persistence. Two near-duplicate repositories.                  |
| kd20072348-arch/NTRO-crypto-tracing                   | 13 commits | An MCP server for conversational crypto tracing. Narrow scope — blockchain only, no stylometry or infrastructure correlation. |
| fayaz01/Dark-Shark                                    | 17 commits | Static <code>index.html</code> plus workflows. No disclosed backend or methodology.                                           |
| BlackCoffeeCode/DarkTrace<br>swaroop-reddy7/Darktrace | Unknown    | No published README detail. Two unrelated projects sharing a name (and both colliding with the commercial Darktrace brand).   |
| Tiruvadhi-Chiruhasini/trace                           | 0 commits  | <b>Repository is empty.</b> Description only.                                                                                 |

# NETRA-X — Position and Actions

## 3.1 Method comparison

| Property                | Noisy-OR            | Weighted sum        | F-S (DRISHTI) | NETRA-X                          |
|-------------------------|---------------------|---------------------|---------------|----------------------------------|
| Parameters derived from | Asserted $w_i$      | Asserted weights    | $m/u$         | $m/u$ , each separately arguable |
| Correlated evidence     | Assumed independent | Assumed independent | Not described | $\lambda = 0.25$ discount        |
| Negative evidence       | Not described       | traceforge only     | Not described | Uncapped, -20 / -15              |
| Saturation behaviour    | → 1.0 with volume   | → 1.0 with volume   | Unbounded     | Capped per family                |
| Per-item explanation    | Hard (product form) | Easy                | Not described | Exact waterfall + artifact hash  |
| Published metrics       | 238 tests           | n = 5 scenarios     | None          | Reproducible bench, seed 42      |

## 3.2 Honest scorecard

### WHERE WE LEAD

- **Dependence discounting** — unique in the field
- **Uncapped contradictions** — only traceforge has anything comparable
- **Family caps that force corroboration** — stylometry 3.0 < threshold 3.73
- **Per-item evidence waterfall** with artifact digests
- **Reproducible benchmark** — DRISHTI publishes none
- **Production frontend** — Next.js + Cytoscape beats Streamlit
- **Retraction over deletion** — nobody else discusses ledger integrity

### WHERE WE TRAIL

- **Tests** — 115 vs 238 (sharan112212)
- **Semantic model** — untrained random weights vs MiniLM / SBERT
- **RBAC** — stored but **never enforced**; two rivals enforce it
- **CI** — slide claims GitHub Actions; no workflow exists
- **Tor** — testbed works but sits outside the repository
- **Stylometry breadth** — ~4 feature types vs 50+

## 3.3 Actions before submission

| Pri | Action                       | Effort   | Why it matters competitively                                                                                                                                                 |
|-----|------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Enforce RBAC                 | ~1 hour  | Column and token claim already exist; only a dependency check is missing. Two rivals enforce it. Cheapest credibility gain available.                                        |
| 2   | Add CI workflow              | ~30 min  | The tech-stack slide shows GitHub Actions and <code>.github/workflows/</code> does not exist. Currently a self-inflicted wound.                                              |
| 3   | Commit the Tor testbed       | ~15 min  | A working hidden service with six planted leaks and 13/13 verification is invisible because <code>testbed/</code> is wholly gitignored. Ignore only <code>.runtime/</code> . |
| 4   | Swap in MiniLM / SBERT       | ~1 day   | Closes the one place a rival is unambiguously ahead, and removes the need to say "untrained" about anything.                                                                 |
| 5   | Wire the isotonic calibrator | ~2 hours | The benchmark fits a calibrator the live API never uses. Closing this makes the published metrics describe the shipped path.                                                 |

| Pri | Action                       | Effort    | Why it matters competitively              |
|-----|------------------------------|-----------|-------------------------------------------|
| 6   | Raise test count toward 150+ | ~half day | Optics against 238. Lower value than 1–3. |

### 3.4 Positioning

IF COMPARED TO DRISHTI (FELLEGI-SUNTER)

"We share the framework. The difference is what we added to it. Fellegi–Sunter assumes conditional independence — so a reused wallet and the co-spend cluster containing it count twice. We discount correlated evidence explicitly within dependence groups. And our benchmark reports numbers: ROC-AUC 0.9028, ECE 0.0044, false-attribution rate zero, reproducible at seed 42."

IF COMPARED TO SHARAN112212 (NOISY-OR)

"Noisy-OR multiplies out complements, which means enough weak signals push the score toward certainty and the weights themselves are asserted rather than derived. Our  $m$  and  $u$  are separately defensible — how often you see a clue between the same operator, versus between strangers. And we can go negative: conflicting PGP keys subtract twenty, uncapped, and reject the link outright."

IF COMPARED TO SENTINEL-X

"Comparable architecture. Ours is auditable end to end — every score decomposes into named evidence items with their discount, their cap and the SHA-256 of the artifact each came from. We also don't claim court admissibility, because admissibility depends on jurisdiction, collection legality and expert testimony, not on a hash chain existing."

THE ONE-LINE SUMMARY

**Sixteen teams are scoring evidence. Fifteen of them count the same mistake more than once.** That is the argument, and it holds against every implementation found.

Sources: GitHub repository search for `SIH26151` and full-text search for "dark web threat actor de-anonymization", September 2026. Each repository was fetched and read individually. Statements of "not disclosed" mean the information is absent from the public README, not that the capability is absent from the code.